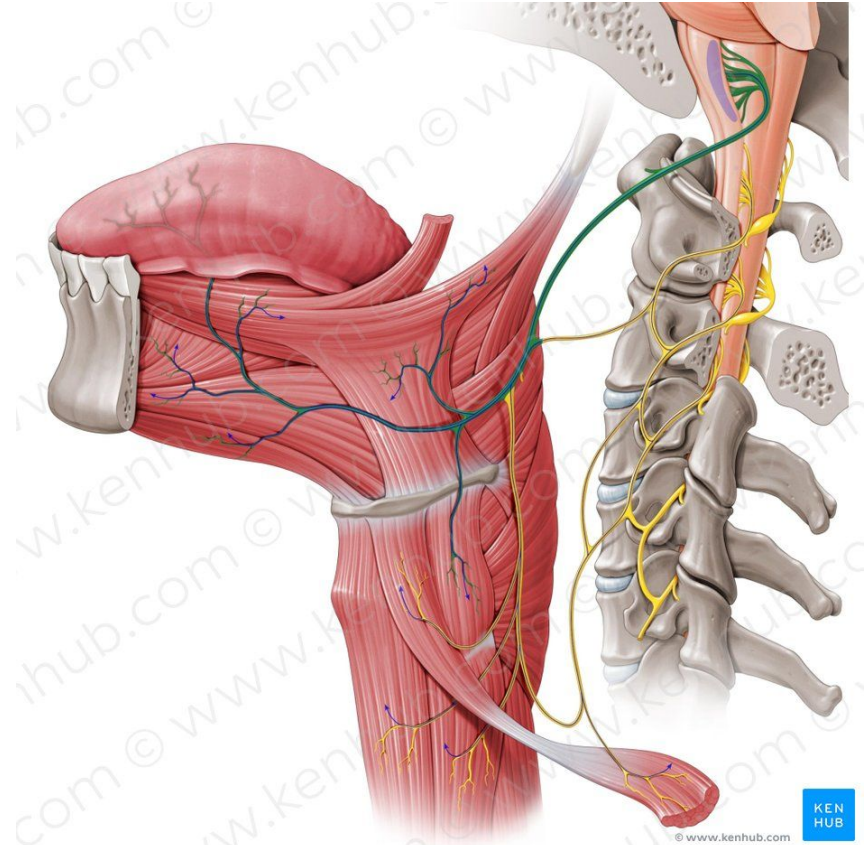


Cranial Nerves 9, 10, 11, & 12

Hypoglossal Nerve (12th):

- It is the 12th cranial nerve.
- It is motor mainly supplies the muscles of tongue.
- **Nucleus:** Hypoglossal nucleus lying in the floor of fourth ventricle below hypoglossal triangle



Floor of the Fourth ventricle

Medullary part

-Inferior fovea

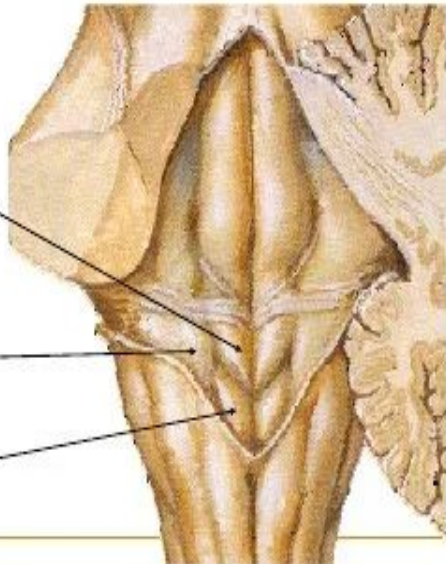
1- Hypoglossal triangle:

overlying hypoglossal nucleus

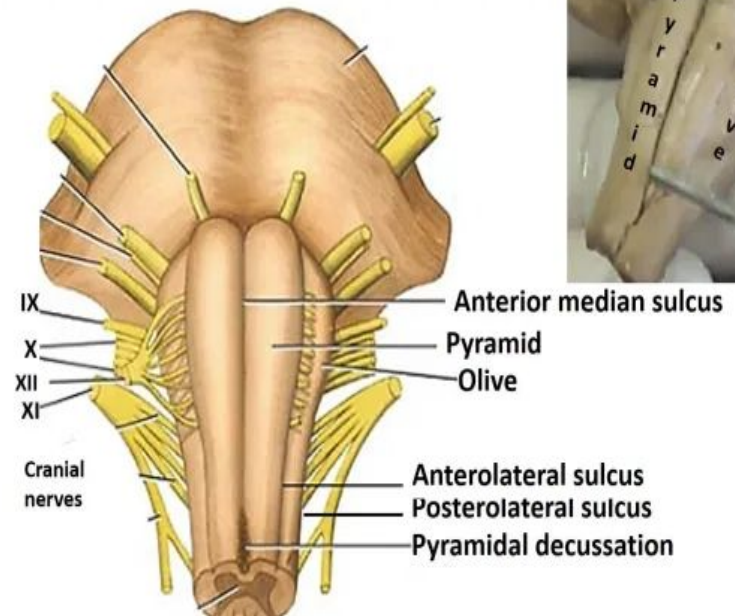
2- Vestibular triangle

3- Vagal triangle:

overlies dorsal nucleus of vagus nerve

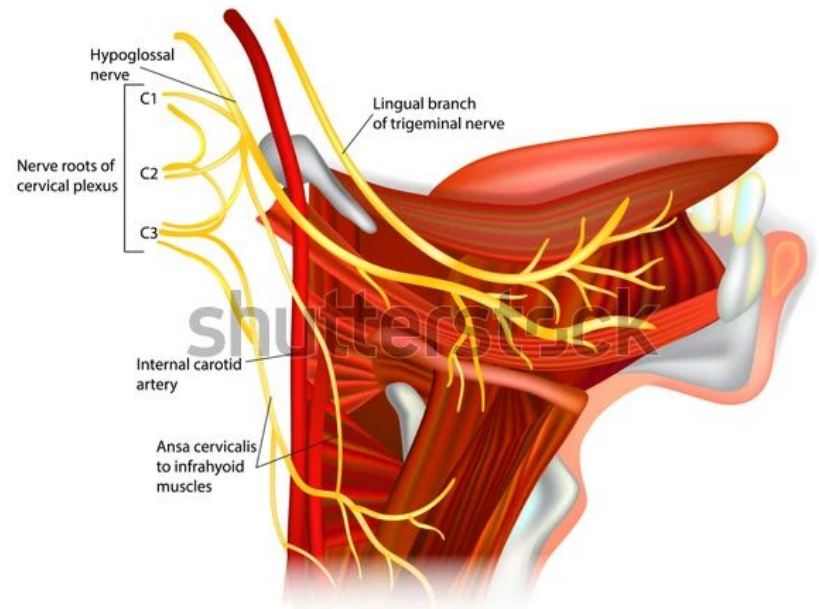


External Features: Ventral Surface of Medulla



Course:

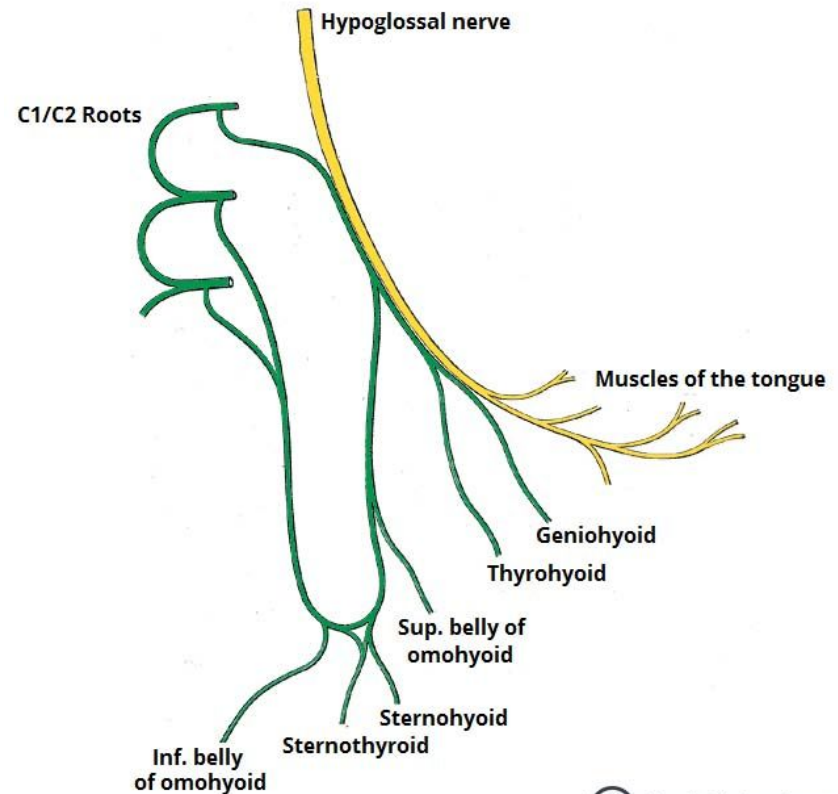
- The nerve rootlets are attached to anterolateral sulcus of medulla and join to form two bundles.
- The nerve exits the cranial cavity through hypoglossal (anterior condylar canal) and descends between internal jugular vein and internal carotid artery deep to parotid gland.
- At the lower border of posterior belly of digastric it curves forwards and enters submandibular region.
- It enters the substance of tongue deep to sub mandibular gland and supplies the muscles of tongue.



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Branches:

- **Hypoglossal nerve proper:** supplies all the intrinsic and extrinsic muscles of tongue except for palatoglossus which is supplied by pharyngeal plexus.
- **Hypoglossal nerve along with C1 nerve:**
 - I. A meningeal branch which supplies bone and meninges of anterior half of posterior cranial fossa.
 - II. Branches are given to thyrohyoid and geniohyoid.
 - III. Upper root of ansa cervicalis.



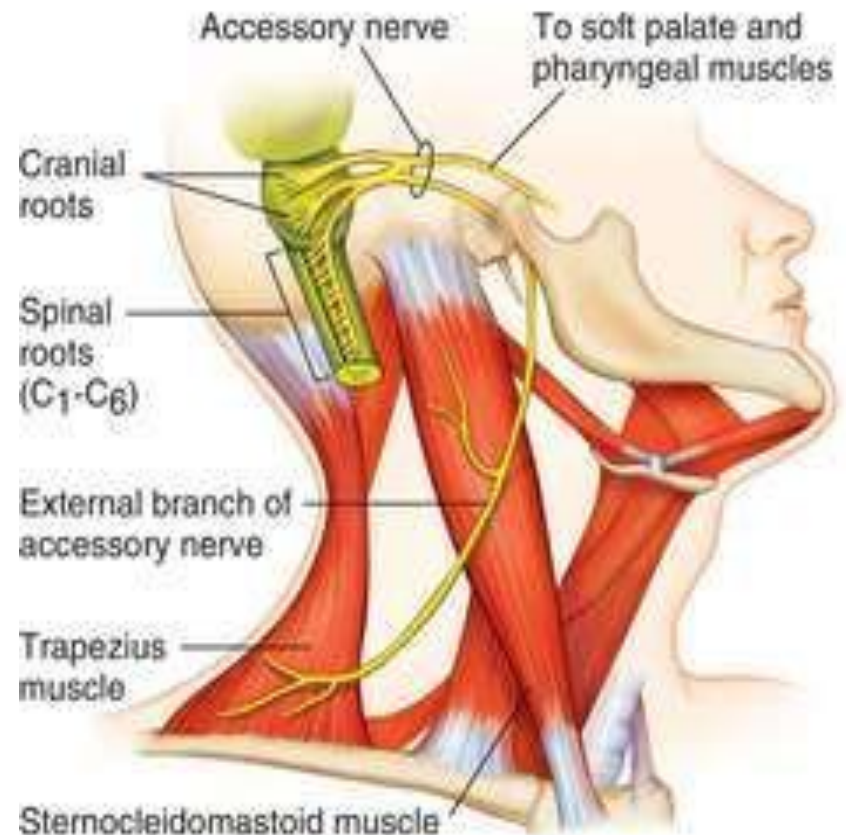
Accessory nerve (11th):

- It is the 11th cranial nerve.
- It is motor and has two roots, cranial and spinal.
- Cranial root is accessory to vagus nerve.
- The cranial root arises from lower part of nucleus ambiguus and the spinal root arises from the long spinal nucleus in the lateral part of anterior gray column extending between segments C1 to C5.



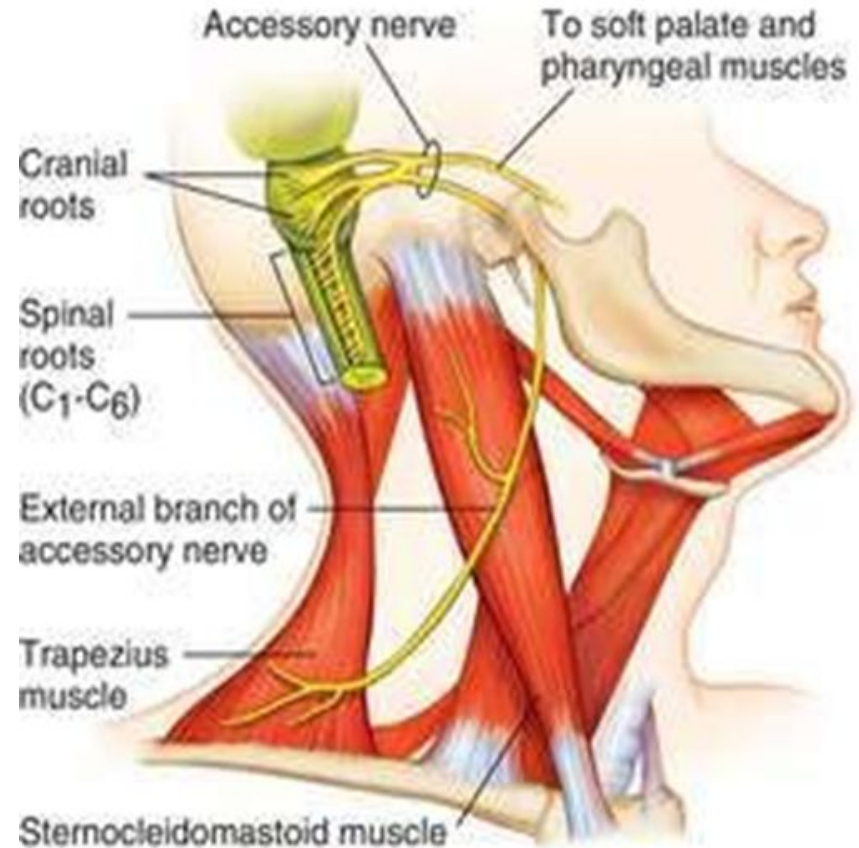
Cranial root/part:

- Cranial part of accessory nerve is attached to posterolateral sulcus by means of rootlets.
- The rootlets unite to form single trunk and it reaches the jugular foramen along with spinal root, glossopharyngeal and vagus nerves.
- In the jugular foramen it unites for a short distance with spinal root and again separates from it after exiting the foramen.
- It then unites with vagus and supplies the muscles of pharynx, larynx and possibly heart through the branches of vagus.



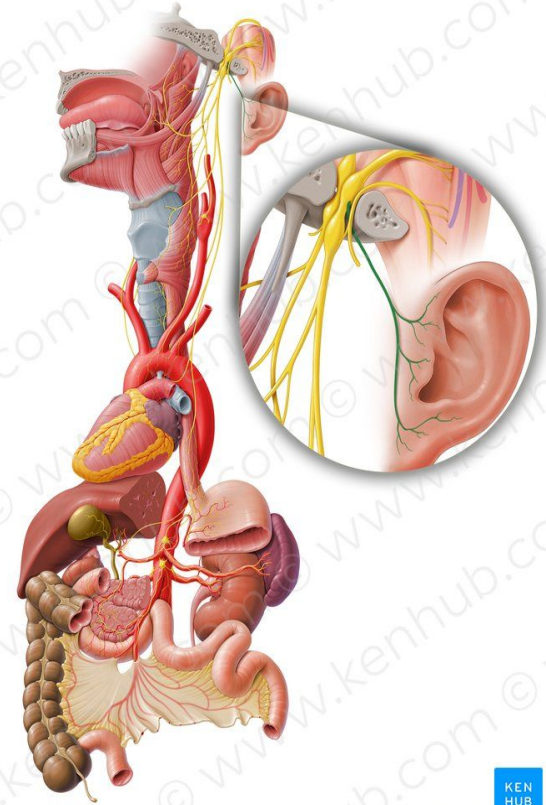
Spinal root/part:

- It arises from upper five segments of spinal cord in the form of row of filaments midway between anterior and posterior nerve roots.
- The row of filaments form a single trunk which ascends in the vertebral canal and enters cranial cavity through foramen magnum.
- Within the cranium it reaches the jugular foramen and unites with the cranial root.
- It separates from the cranial part soon after exiting the jugular foramen.
- It descends vertically and pierce the anterior border of sternocleidomastoid and communicates with C2 and C3 nerves.
- It arises from posterior border of sternocleidomastoid and runs in the roof of posterior triangle.
- It ends by supplying the trapezius, here it communicates with the C3 and C4 nerves deep to it.



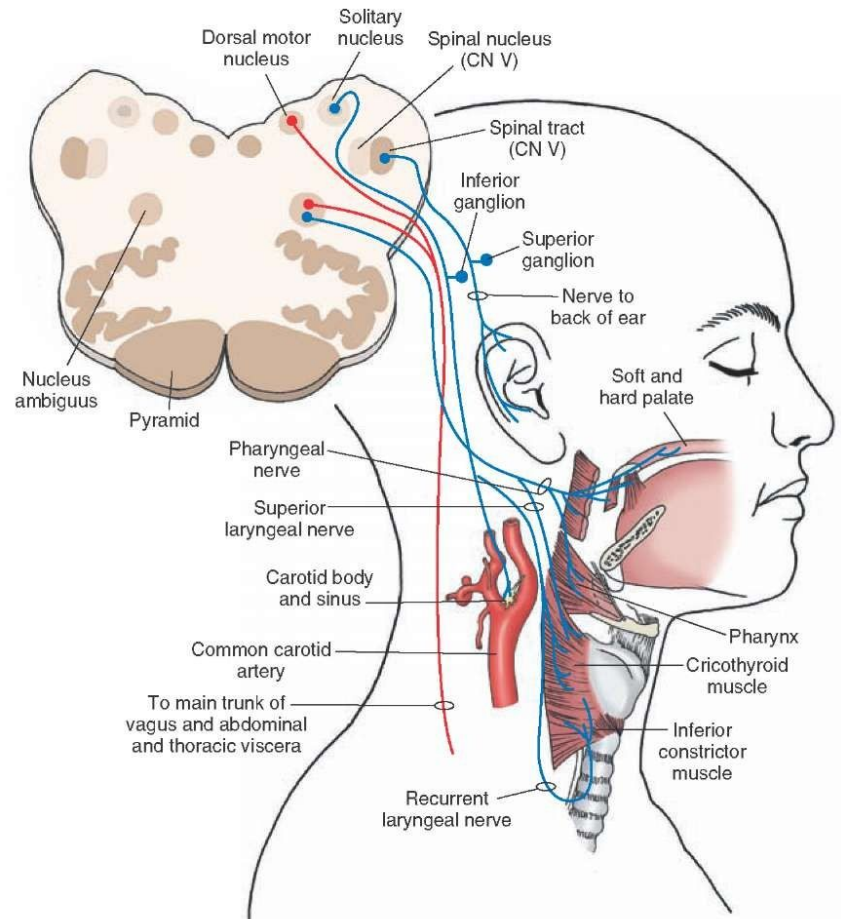
Vagus nerve (10th):

- It is the 10th cranial nerve.
- It is mixed and named so (vagus) due to its extensive course throughout the head, neck, thorax and abdomen.
- It has two ganglia: the superior ganglion which lies in jugular foramen and inferior ganglion which lies near the base of skull.



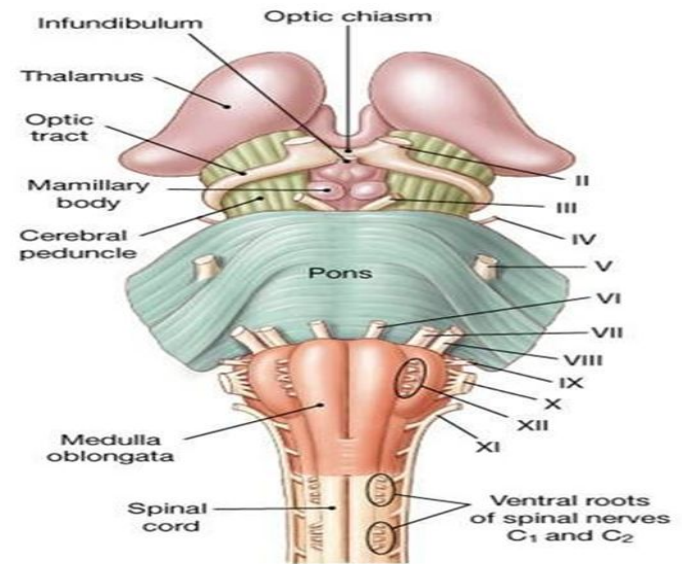
Nuclei:

- 1) Nucleus ambiguus (motor)
- 2) Dorsal nucleus of vagus (parasympathetic)
- 3) Nucleus of tractus solitarius
- 4) Nucleus of spinal tract of trigeminal nerve

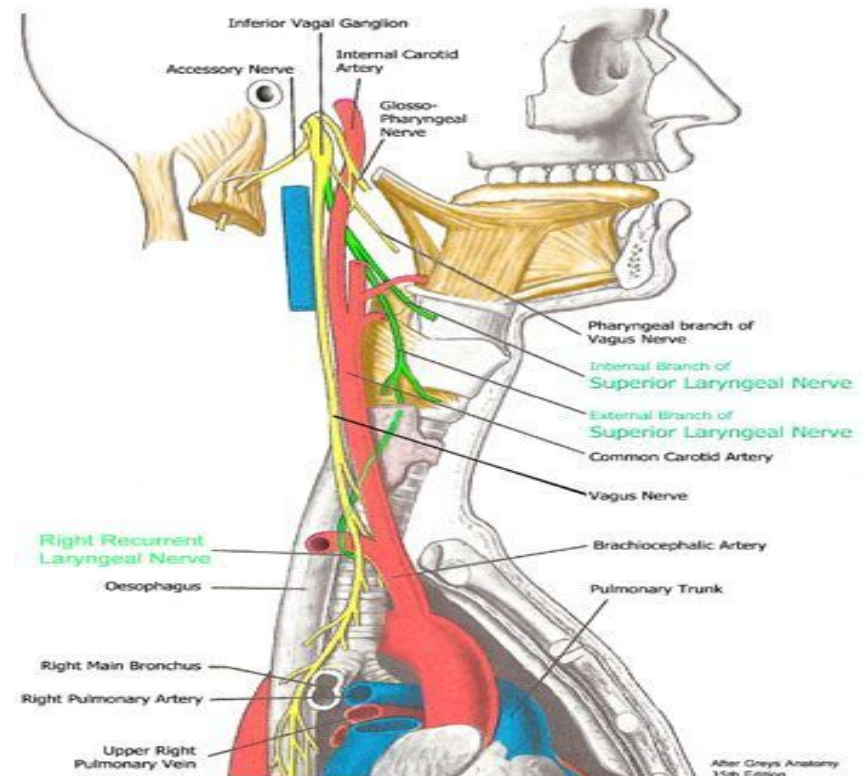


Course:

- The nerve is attached to posterolateral sulcus of medulla by means of rootlets which unite to form a single trunk.
- The nerve then reaches the jugular foramen along with glossopharyngeal and cranial part of accessory nerve.
- After exiting the cranial cavity through jugular foramen, the nerve descends within carotid sheath posterior to internal jugular vein and common and internal carotid arteries.
- Both right and left vagi reach the root of the neck, the right vagus enters thorax crossing subclavian artery and the left vagus enters thorax passing behind internal jugular vein.



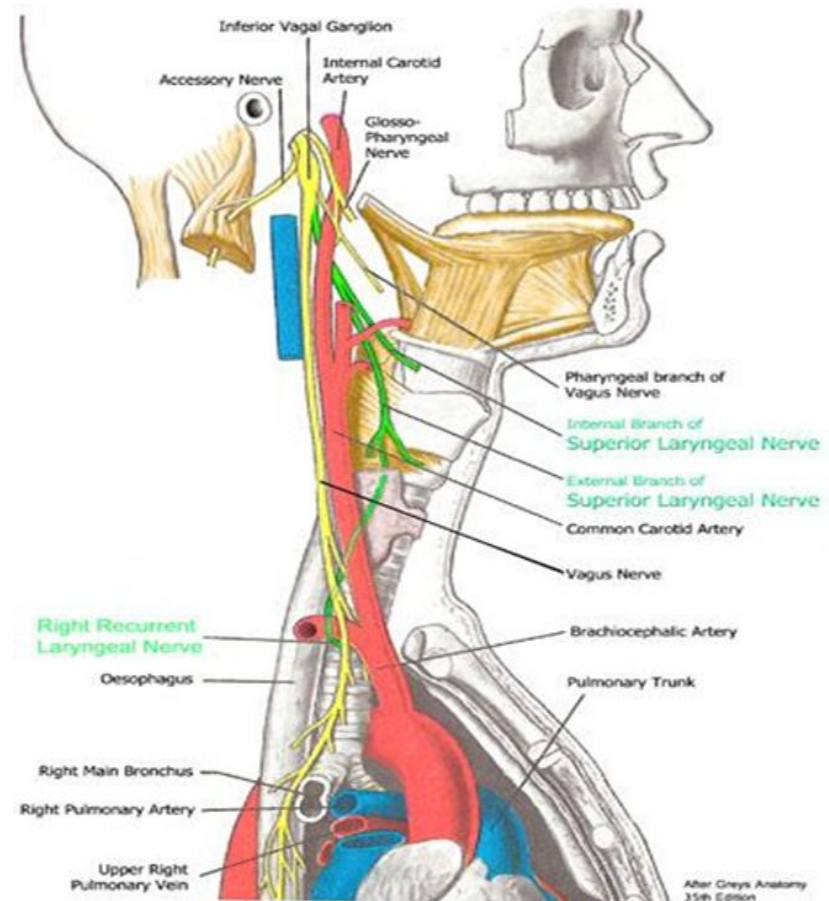
(b) Anterior view



Branches:

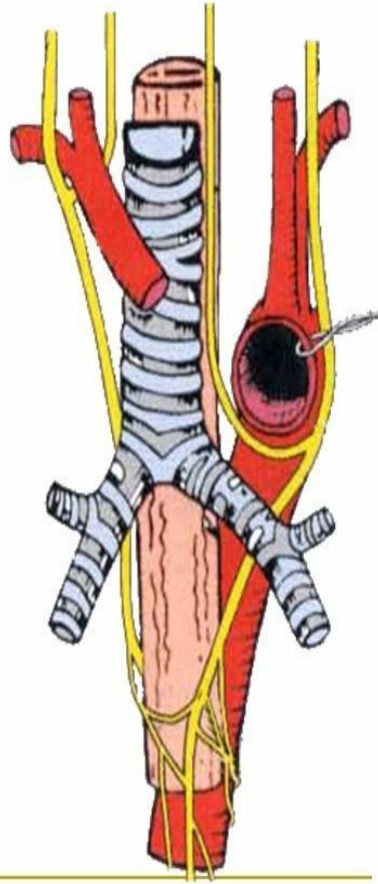
- **In Head:**
 - I. Meningeal branch
 - II. Auricular branch
 - III. Communicating branches to glossopharyngeal and cranial part of accessory nerve
- **In Neck:**
 - I. Pharyngeal branch
 - II. Superior laryngeal branch which divides into external and internal laryngeal nerves
 - III. Right recurrent laryngeal branch (as left recurrent laryngeal branch arises in thorax)
 - IV. Cardiac branches

- **N.B:** external laryngeal nerve accompanies superior thyroid artery and recurrent laryngeal nerve is related to inferior thyroid artery.

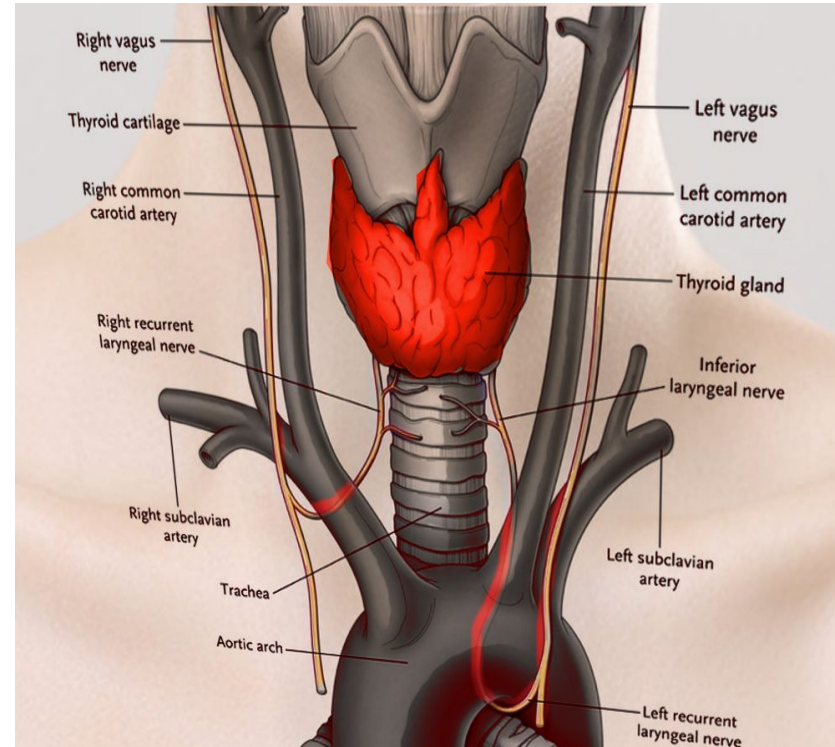


Recurrent laryngeal nerves

- Right one hooks around right subclavian artery, left one hooks aortic arch
- Both ascend in tracheo-esophageal groove
- Nerves enter larynx posterior to cricothyroid joint, the nerve is now called inferior laryngeal nerve
- Innervations: laryngeal mucosa below fissure of glottis , all laryngeal laryngeal muscles except cricothyroid

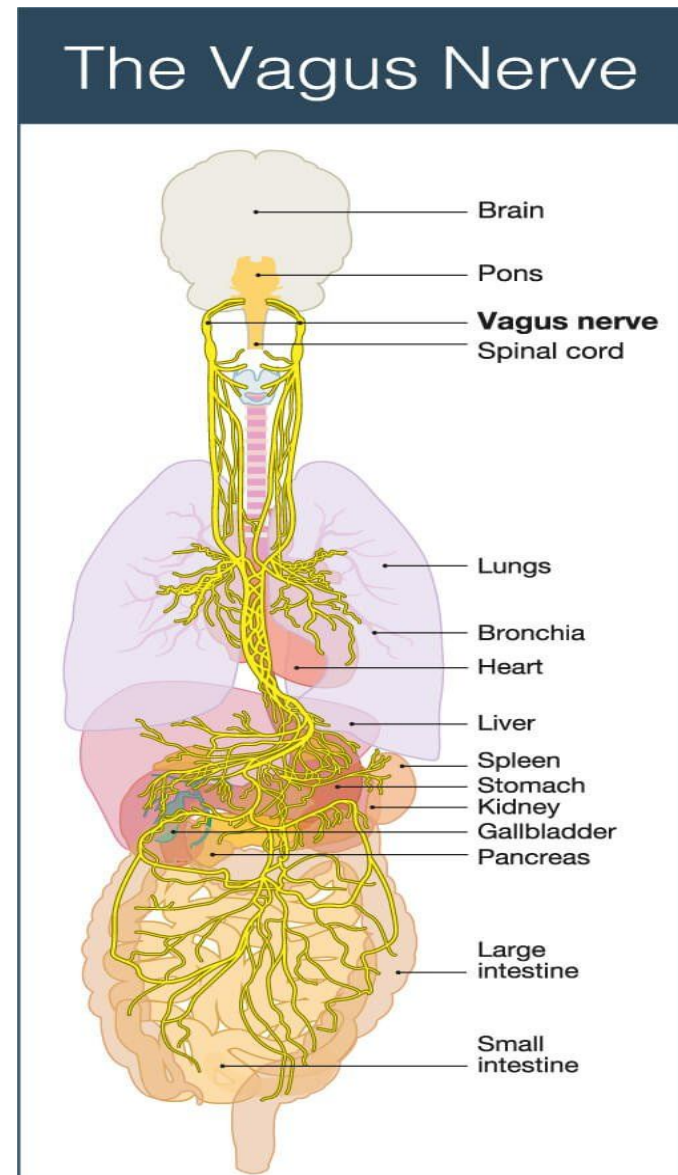


Bronchial and esophageal branches



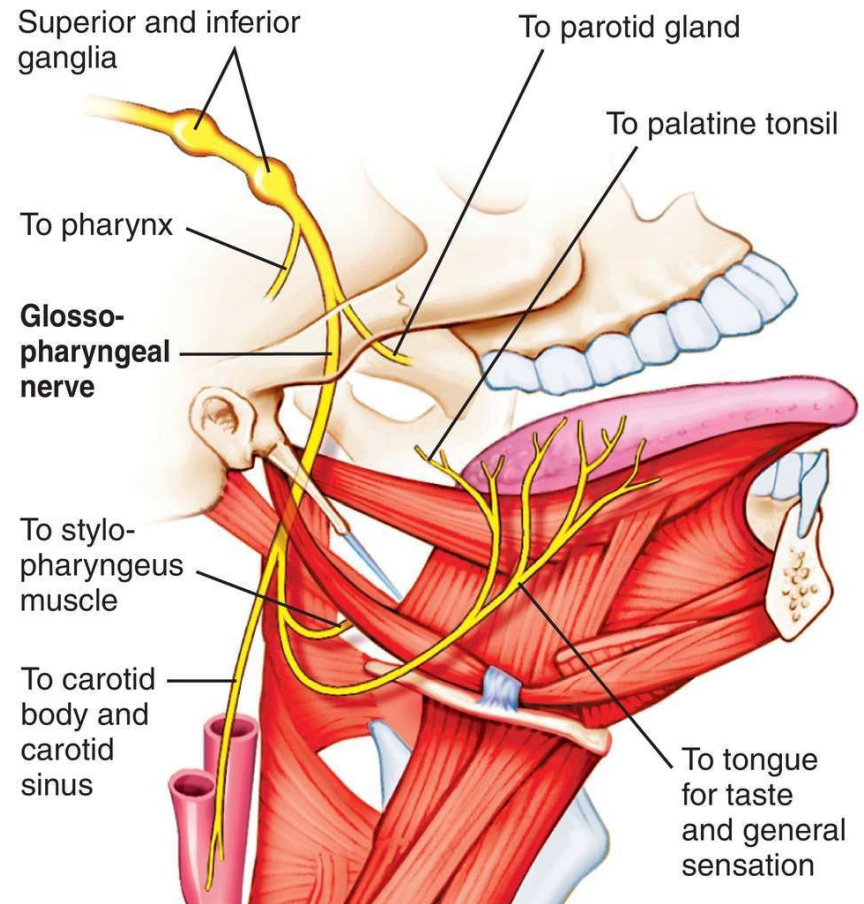
Distribution of Vagus nerve in Thorax & Abdomen:

- Below the neck, the vagus also supplies parasympathetic fibers to thoracic viscera and most of the GIT from the esophagus up to left colic flexure.
- It also supplies the accessory glands of GIT such as liver & pancreas.



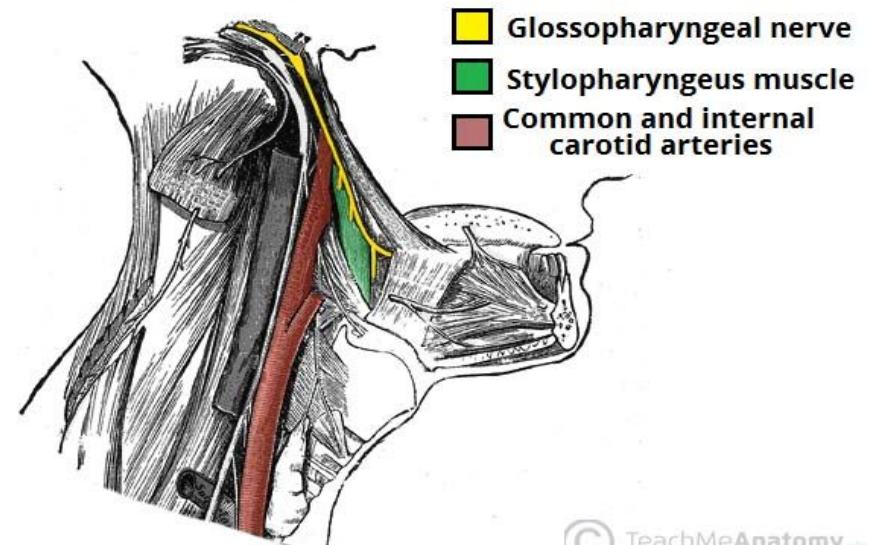
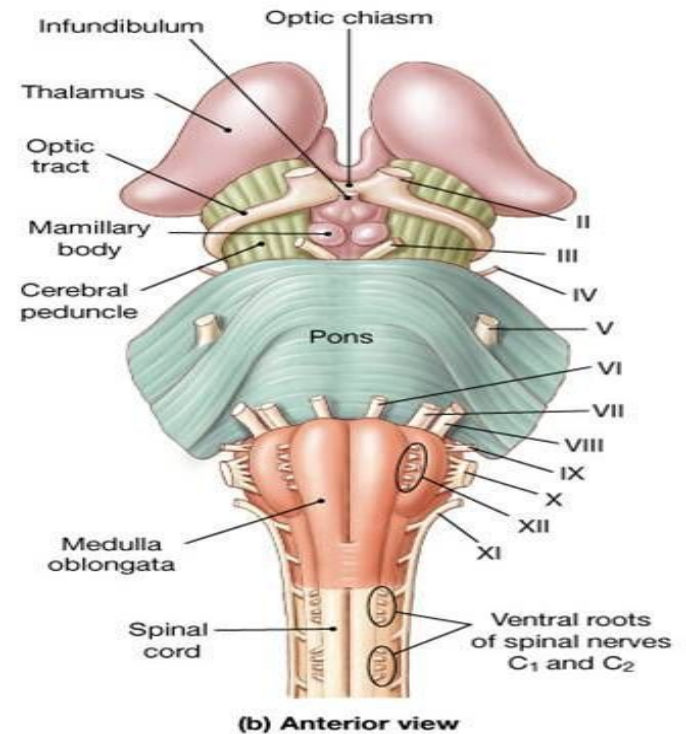
Glossopharyngeal nerve (9th):

- It is the 9th cranial nerve.
- It is mixed and the nerve of third pharyngeal arch.
- It is motor to stylopharyngeus muscle and secretomotor to parotid gland.
- It is gustatory to posterior one-third of the tongue including circumvallate papillae and sensory to pharynx, tonsil, posterior one-third of tongue, carotid sinus and body.



Course:

- The nerve is attached to posterolateral sulcus of medulla via 3 to 4 rootlets just above vagus.
- The rootlets unite to form single trunk which leaves the cranial cavity through jugular foramen.
- The nerve descends between internal jugular vein and internal carotid artery and winds round laterally the stylopharyngeus to reach the pharynx.
- It passes between external and internal carotid arteries to enter submandibular triangle where it ends by splitting into tonsillar and lingual branches.
- It has superior and inferior ganglia at the base of skull.



Branches:

- Tympanic nerve (takes part in the formation of tympanic plexus)
- Carotid branch (supplies carotid sinus and body)
- Pharyngeal branch (contributes in the formation of pharyngeal plexus)
- Muscular branch (for stylopharyngeus)
- Tonsillar branches
- Lingual branches (for taste and general sensations from posterior one-third of tongue including circumvallate papillae)

